MSG No. 128.404 $P4'/m'n'c$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4'/m'..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,4,5,6,9,10,13,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3,7,8,11,12,14,15]`

Table 2: Wyckoff site: `2b`, site symmetry: `4'/m'..`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,4,5,6,9,10,13,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[2,3,7,8,11,12,14,15]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,4,13,16]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,3,14,15]`
3	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[5,6,9,10]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[7,8,11,12]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,4,11,12]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[2,3,9,10]`
3	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[5,6,14,15]`
4	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[7,8,13,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `4'..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4,9,10]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[2,3,11,12]`
3	$[0,\ 0,\ -z]$	`[5,6,13,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8,14,15]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[2,3]`
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[5,6]`

continued ...

Table 6

No.	position	mapping
4	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[7,8]
5	$\left[\frac{1}{2},\ 0,\ z\right]$	[9,10]
6	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[11,12]
7	$\left[0,\ \frac{1}{2},\ -z\right]$	[13,16]
8	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[14,15]

Table 7: Wyckoff site: 8g, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[1,11]
2	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[2,10]
3	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	[3,9]
4	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[4,12]
5	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[5,15]
6	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	[6,14]
7	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[7,13]
8	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[8,16]

Table 8: Wyckoff site: 8h, site symmetry: m'..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,16]
2	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[2,15]
3	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[3,14]
4	$[-x,\ -y,\ 0]$	[4,13]
5	$[y,\ -x,\ 0]$	[5,10]
6	$[-y,\ x,\ 0]$	[6,9]
7	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[7,12]
8	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[8,11]

Table 9: Wyckoff site: 16i, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[2]
3	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$[y,\ -x,\ -z]$	[5]
6	$[-y,\ x,\ -z]$	[6]
7	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[7]
8	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[8]
9	$[-y,\ x,\ z]$	[9]
10	$[y,\ -x,\ z]$	[10]

continued ...

Table 9

No.	position	mapping
11	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[12]
13	$[-x,\, -y,\, -z]$	[13]
14	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[14]
15	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[15]
16	$[x,\, y,\, -z]$	[16]